# Lab 03: Chaotic Text Generator

**Author: Ayoub El Houani** One-page project report

## 1. Objective and method

This project compares a plain part-of-speech HMM, a role-split HMM and a trigram generator. A Streamlit application generates batches, displays tags, computes metrics, exports CSV and tags user input with log-space Viterbi. The synthetic corpus is split by document into 960 training and 240 test sentences. Its small grammar makes this a teaching experiment, not a measure of real English quality.

## 2. Generation results at $T = 1.0$

Each model generated 500 sentences with seed 42. *Valid* means the tag pattern was seen in training; *plausible* means the first noun–verb pair was seen; *novel* means the full sentence was not copied; distinct-2 is the fraction of unique word bigrams.

| Model | Valid pattern | Plausible pair | Novel | Distinct-2 |
|---|---|---|---|---|
| Plain HMM | 16.2% | 14.8% | 99.8% | 0.198 |
| Role-split HMM | 100.0% | 60.6% | 99.4% | 0.148 |
| Trigram | 57.0% | 76.0% | 91.2% | 0.105 |

**Interpretation.** The plain HMM reuses POS states at different sentence positions, so it can combine incompatible transitions. Role-split states retain subject, object or prepositional roles. They produce 100% valid patterns in this sample, but do not learn meaning or guarantee grammatical English. Trigram tags are estimated with the role-split HMM.

## 3. Settings and generated examples

For a **writing toy**, $T = 1.5$ is a reasonable choice for variety; for **autocomplete**, $T = 0.5$ favours common continuations. These are heuristics, not measured autocomplete results.

With the role-split HMM, seed 42 and $T = 1$, “the children eat the bread at the park .” is a plausible novel sentence. “the farmers eat the river by the school .” has a valid structure but fails in **meaning**: independent word emissions do not model semantic compatibility. Its noun–verb pair was seen, showing the limits of the plausibility proxy.

## 4. Evaluation on real text

Both HMMs achieved 100% token accuracy on the 240 held-out synthetic sentences. On UD English EWT (12,544 training and 2,077 test sentences; CC BY-SA 4.0), the measured results were:

| Tagger | Accuracy | Known | Unknown | Whole sentence |
|---|---|---|---|---|
| Most-frequent tag | 86.2% | 90.8% | 28.6% | 31.9% |
| HMM + Viterbi | 88.0% | 92.9% | 28.2% | 40.4% |

Unknown words make up 7.5% of test tokens. Real web text has 17 tags, more varied patterns, ambiguity and unseen words; the model’s smoothed emissions lack word-shape information for unknowns. The synthetic grammar is much smaller and more regular. Results can vary with treebank revisions.